

Tutorial Sheet - II

1. Are the vectors $a_1 = (-1, 1, 2, 4)$, $a_2 = (2, -1, -5, 2)$, $a_3 = (1, -1, -4, 0)$, $a_4 = (2, 1, 1, 6)$ linearly dependent in $\mathbb{R}^4$?

2. Let V be the vector space of all 2×2 matrices over $\mathbb{C}$.

(i) Prove that V has dimension 4 by exhibiting a basis for V which has four elements.

(ii) Let $W = \{A \in M_{2 \times 2}(\mathbb{C}) : A_{11} + A_{22} = 0\}$.

Show that W is a subspace of V and find the dimension of W .

3. Let X be the vector space of all polynomials $p(t)$ of degree $\leq n$, and let Y be the set of polynomials in X that vanishes at $t_1, \dots, t_j$, $j \leq n$.

(i) Show that Y is a subspace of X .

(ii) Find $\dim X$ and $\dim Y$.

(iii) Find $\dim(X/Y)$.

4. Let V be a vector space and A be a linearly independent subset of V . Prove that A is a basis for V if and only if it is a maximal linearly independent subset of V , that is addition of any vector to A will result in a linearly dependent set.

5. Let V be a vector space, and let $\text{Span}(A) = V$, for a subset A of V .

Prove that A is a basis for V if and only if it is a minimal spanning set, that is removal of any vector from A will result in a set which does not span V .